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Title of the Research Study	FIELD-LABEL DISAMBIGUATION AND COMPLETION-TIME REDUCTION IN PDF FORMS USING LAYOUTLMV3 WITH GRAPH NEURAL NETWORK ARCHITECTURE
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Background	Participants in this study will be recruited through purposive sampling. Recruitment will be carried out by posting announcements within university classes, online communities, and professional networks, as well as by reaching out to individuals whose work involves regular interaction with PDF forms, such as virtual assistants, office staff, and students. The recruitment process will emphasize voluntary participation, with interested individuals invited to contact the research team for scheduling. Prior to participation, prospective respondents will be screened based on the inclusion criteria of being at least 18 years old, computer literate, and having prior experience in filling out PDF forms. Those who meet the criteria will be invited to participate in the study sessions, where informed consent will be obtained before any research activity begins. No compensation or incentives will be provided, and participants will be reminded that they may withdraw from the study at any point without penalty.
Purpose of the Study	The purpose of this study is to evaluate the effectiveness of the proposed system called <i>IntelliForm</i> , which is designed to help people fill out PDF forms more accurately and efficiently. Many forms use labels such as “Date” or “Name” that can be unclear, repeated, or placed in different areas of a page, which makes them confusing or time-consuming to complete. IntelliForm uses artificial intelligence, specifically a document-understanding model enhanced with a graph neural network, to identify these form labels automatically

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	and generate short, descriptive summaries to guide users in filling out forms. The study aims to test whether the tool can correctly identify field labels, provide understandable descriptions, and reduce the time needed to complete forms compared to manual filling and an existing baseline model. Data collected in this study will be used solely for the completion of the researchers' undergraduate thesis in the Bachelor of Science in Computer Science program at the Polytechnic University of the Philippines. The researchers declare that there are no financial or personal conflicts of interest related to this study.
Procedures of the Study	Participants will be recruited through purposive sampling, focusing on adults who have prior experience in filling out PDF forms, such as students, professionals, administrative staff, and virtual assistants. Interested individuals who meet the inclusion criteria will be invited to scheduled sessions. Upon arrival, they will first receive a 15-minute orientation explaining the study's objectives, procedures, expected time commitment, and their rights as participants. Written informed consent will then be obtained, with names and signatures recorded for validation purposes. Each participant will also be assigned a unique code (e.g., P001) to anonymize their performance data and maintain confidentiality. Following consent, participants will be asked to complete form-filling tasks under three different conditions: (1) manual completion without system assistance, (2) baseline completion using vanilla LayoutLMv3, and (3) assisted completion using IntelliForm, the proposed system. The order of these conditions will be randomized to avoid learning effects or order bias. The activity will involve publicly available PDF forms drawn from domains such as government, education, healthcare, human resources, and real estate. To safeguard participant privacy, they will be instructed not to provide real personal or sensitive details when filling out the forms. Instead, only dummy or placeholder data (e.g., "Juan Dela Cruz," "123-456-789") will be used to simulate entries. Each session will last approximately 20 to 30 minutes, depending on the participant's pace. Throughout the session,

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	task completion times will be automatically recorded by an inference profiler, and the system will log metrics related to field detection accuracy and summary generation. No actual form content or personal data will be stored. Only performance-related data such as timestamps, accuracy scores, and system outputs will be retained. All logged data will be anonymized through participant codes and stored securely for later statistical analysis.
Benefits	The study offers indirect benefits to participants by exposing them to an innovative AI-based tool for completing PDF forms, which may increase their awareness of how emerging technologies can simplify everyday digital tasks. By interacting with IntelliForm, participants may gain insights into the future of intelligent systems that improve productivity and reduce repetitive work. Beyond individual benefits, the research contributes to the field of computer science by introducing and testing a novel method for disambiguating field labels and reducing completion time in PDF forms. The findings may inform the design of future digital tools that streamline administrative workflows in education, healthcare, government, and business settings. In the long term, the results of this study may help reduce errors, improve efficiency, and enhance user experience in form-filling tasks, ultimately benefiting institutions, professionals, and the public.
Risks	This study poses no anticipated physical, psychological, social, or legal risks to the participants. The tasks required, primarily filling out sample PDF forms using the IntelliForm system and other baseline methods, are similar to ordinary form-filling activities and do not introduce harm or sensitive data exposure. The only possible inconvenience may be the time and effort required to complete the tasks. To mitigate this, the researchers will ensure that participants are fully briefed beforehand, provided with clear instructions, and given sufficient time to complete the activities at their own pace. Participation is voluntary, and participants may withdraw at any point if they feel uncomfortable, without penalty.

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	take part in the study, and they may withdraw their participation at any time without facing any negative consequences. Should a participant choose to withdraw, they may also request the removal of their data from the study records, provided that the request is made before the data has been consolidated and analyzed. To withdraw, participants may communicate directly with the researchers through the contact number or email addresses indicated in this form. Once data analysis has begun and individual responses can no longer be separated, withdrawal of data will no longer be possible.
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Contact Information:

If you have questions about this research study, or you experience adverse effects because of participating in this study, you may contact the researcher/s whose contact information is provided on the first page.

The plan for this research study has been reviewed by the PUP Research Ethics Committee (PUP-REC). If you have questions regarding your rights as a research participant, or if problems arise that you do not feel you can discuss with the researcher/s, please call us at (02) 5335-1787 local 235.

Consent Statement:

I have read and understood the provided information and have had the opportunity to ask questions. I understand that my participation is voluntary and that I am free to withdraw at any time, without giving a reason and without cost. I understand that I will be given a copy of this consent form. I voluntarily agree to take part in this study.

Signature over Printed Name of Participant

Date

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December 2, 2025
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